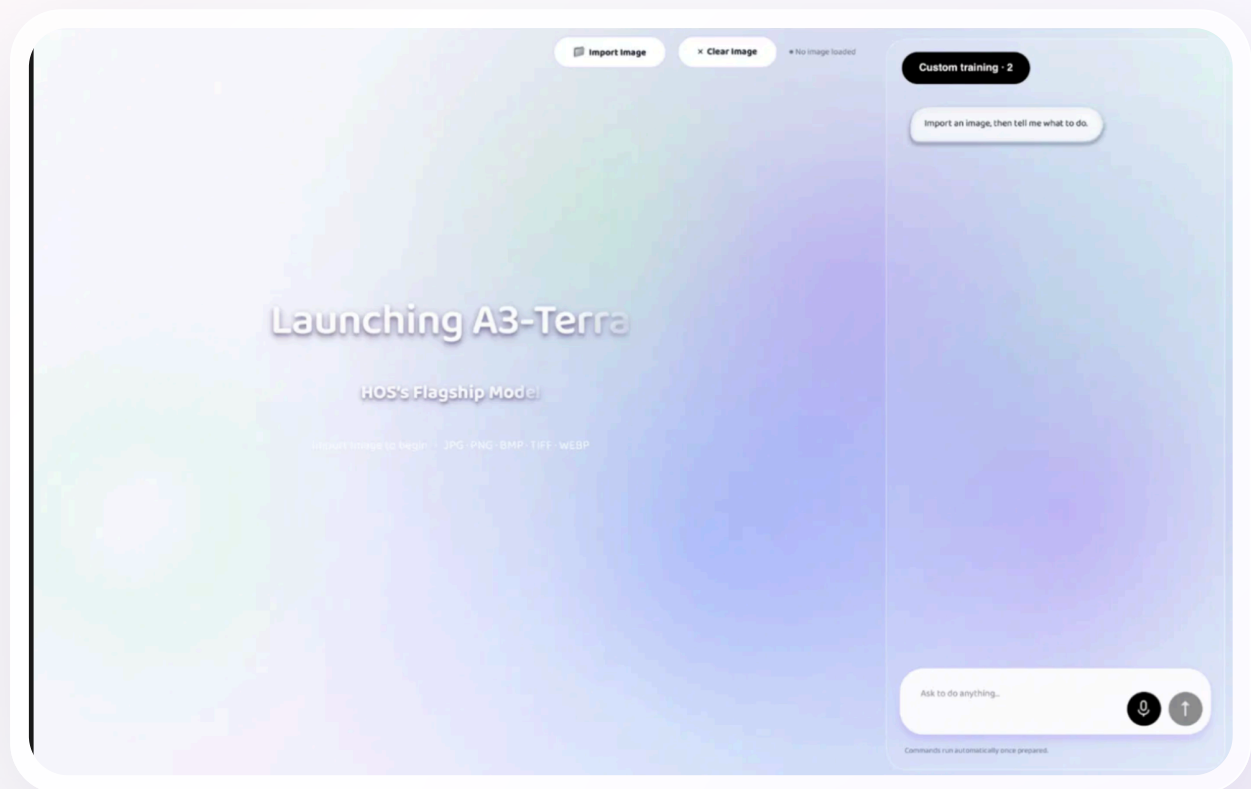


From image to action.

A practical guide to A3-Terra, the desktop operating surface for understanding a physical board, planning a robot task, simulating it, and sending commands to hardware.



Homepage image: authentic A3-Terra launch screen supplied for this guide.

01

What A3-Terra does

The basic operating model.

A3-Terra is the Humanoid Operating System flagship desktop surface. It uses a fixed **20 × 11 working board**, with columns A–T. A scene image plus an operator instruction becomes a coordinate-aware action plan, which can be simulated and optionally transmitted over USB when the hardware is armed.

01

Bring in a scene

Import JPG, PNG, BMP, TIFF or WEBP, load an example, or connect a USB camera.

02

Understand the board

Vision outlines discrete objects and maps them onto the 20 × 11 grid.

03

Ask for work

Type or dictate a household-robot task in plain language.

04

Review and act

Watch simulation first, then transmit only when hardware is calibrated and armed.

Safe first habit

Simulation is the default rehearsal. Connect and arm USB hardware only after AprilTag calibration and after the command sequence looks correct in simulation.

02

Get ready in five minutes

A clean first-run sequence.

01

Launch

Main canvas on the left, AI sidebar on the right. An empty board shows the launch state over the animated wallpaper.

02

Add API key

Settings → API Config → Add manual API key. The key is saved in HOS data/api_key.json.

03

Pick a scene

File → Import Image, Insert → Pictures, an example with Ctrl+E, or a live USB camera.

04

Check then task

Show the full board, use even lighting, then try a simple task and observe the simulation.

API key requirement

Model-driven work requires a usable OpenAI API key. A3-Terra does not embed the key in its source.

Scenes, views & the 20 × 11 board

Give the planner enough visual information to work from.

TOP

Top view

Primary reference for board coordinates, placement, object footprints and wide contact passes.

ISOMETRIC

Isometric view

Useful for depth, height, overlap, stacking and orientation when a top view is insufficient.

SIDE

Side view

Useful for vertical structure, handles, lids, openings and height-sensitive grasp planning.

Image quality checklist

- Show the entire 20 × 11 working area in frame.
- Use even lighting and avoid strong directional shadows.
- Keep the camera steady. Motion blur reduces outline accuracy.
- Avoid reflections and glare on knives, metal bowls and shiny plastic.
- Use higher resolution for thin or overlapping objects.

Detection defaults

SETTING	DEFAULT	PURPOSE
TOUCH_THRESHOLD	0.15	Fraction of a cell an object must cover to claim it.
SMALL_OBJ_CELLS	1.5	Area threshold for sub-cell objects.
SMALL_TOUCH_THR	0.45	Stricter coverage for small objects.

SETTING	DEFAULT	PURPOSE
PADDING_KEEP_MIN	0.55	Requires enough polygon area to remain inside the real photo.

Task workflow

Ask naturally. Let A3-Terra plan precisely.

A

Describe

State what to move, clean, open or place. Be concrete.

B

Clarity check

Real ambiguity produces a question. Resolved scenes proceed without needless clarification.

C

Read the plan

The generated sequence appears in chat with meaningful stages.

D

Execute deliberately

Simulate first. Then calibrate, connect, arm and send.

TIP

Use scene names

Say “the red cup” or “the knife next to the carrot” when needed.

Good task language

- “Move the bowl to an empty area.”
- “Pick up the knife by its handle.”
- “Clean the table with the cloth.”

If it asks a question

Answer in plain language. Repeated clarification usually means the scene image is insufficient, so add another view or re-import a clearer photo.

05

Planning pipeline, memory & custom training

The source separates the major reasoning stages.

01

Vision

Identifies discrete physical objects, outlines them and maps positions to the board.

02

Components

Can inspect a crop for actionable parts such as handles, rims or controls.

03

Dexterity

Checks whether the requested plan is feasible for the robot's constraints.

04

Gripper AI

Suggests safer grip cells, such as a knife handle or plate rim.

05

Planner

Writes the board-aware action sequence and reports missing objects instead of inventing locations.

Planner guardrail

When an object is not on the board, the planner reports the absence and skips that sub-task rather than fabricating a location.

Memory

After a task, the memory check can propose a standing instruction. You choose whether to save it.

Custom instructions

Instructions layer onto the default rules and are stored in **HOS data/custom_instructions.json**. Keep them specific and conditional.

Examples from the source

```
When the task involves a knife, always request grip on the handle, never the blade. If two bowls are present, prefer the larger one unless the user names a colour.
```

Voice, video, simulation & calibration

Fast input, reviewable output and controlled rehearsal.

Voice dictation

Use the microphone to dictate. On macOS, the right Option key can dictate and send in one action. Clean Up Dictation performs a second pass.

Video to instruction

A3-Terra extracts audio from supported video or audio/video files, transcribes it and presents the text for review.

Simulation

Playback can be adjusted from 0.5× to 4×. Stop, re-run and object-label controls are available.

Before trusting a plan

- Does each outline contain the intended object?
- Does the motion path cover the intended cells?
- Is the grip point plausible?
- Is the destination clear and reachable?
- Are waits and tool changes sensible?
- Are follow-up movements correct?

Calibration before hardware

Use Extensions → Hardware Connect → AprilTag Calibration before real-world execution. Re-calibrate after any camera or board movement.

Hardware: connect intentionally

Generating a plan and sending commands are separate states.

01

Attach

Connect the robot/controller and, if needed, a USB camera.

02

Select

Choose the serial port and matching baud rate.

03

Calibrate

Confirm the camera view and complete AprilTag calibration.

04

Arm

Enable Send Commands over USB only when ready for a validated plan.

Safety interlock

A connected serial port is not enough. USB command sending must also be armed. Never leave arming enabled between sessions.

Common hardware failures

PROBLEM	CHECK
USB commands ignored	Port is connected but Send Commands over USB is not armed.
Outlines drift	Physical board or camera moved. Re-run AprilTag calibration.
No serial devices	Refresh ports, check cable/driver and select a port before connecting.
Wrong baud / garbled data	Match the controller baud rate before connecting.
Camera not listed	Use Camera → Connect USB Camera and close other apps using it.

Settings, models, shortcuts & HOS data

Configuration is exposed rather than hidden in the source.

Simulation

Playback speed, wait cap, second-pass verification, snap-to-pixels, Gripper AI, verbose output and object labels.

Models

Vision, Dexterity, Clarity, Memory, Planner, Dictation tidy and Speech to text are independently editable.

Detection

Touch threshold, max touch cells, padding keep minimum and background detection controls.

Network

Request timeout, retry count and retry backoff.

Model defaults from A3-Terra.py

STAGE	DEFAULT
Vision	gpt-5.4
Planner	gpt-5.6-terra
Dexterity / Clarity / Memory	gpt-5.4-mini
Voice tidy	gpt-5.4-nano
Speech	gpt-4o-transcribe

Ctrl+I

Import

Import an image.

Ctrl+E

Examples

Open an example scene.

F11

Fullscreen

Enter or leave fullscreen.

Operating checklist & troubleshooting

Use this before every physical run.

Recommended chain

01

Clear scene

02

Check
outlines

03

Set task

04

Inspect plan

05

Simulate

06

Calibrate

07

Connect

08

Arm

09

Supervise

Fast recovery

SYMPTOM	RECOVERY
Missing object	Improve lighting, add a view, or re-import a clearer photo.
Grip on wrong part	Improve the handle cue, enable Gripper AI, or add a narrow custom instruction.
Plan invents a location	This should not happen. Treat it as a guardrail failure.
Repeated clarity questions	Add a complementary view or use a clearer scene image.
Network/model error	Review API key, timeout, retry count and backoff.

Best practice

Always start with simulation. After changing model IDs or custom instructions, run an example scene first.

Quick reference

The commands and locations you will use most often.

ACTION	WHERE
Import image	File → Import Image or Ctrl+I
Clear image	File / toolbar → Clear Image
Open examples	Examples menu or Ctrl+E
Custom training	Insert → Custom Training
Manage views	Insert → Pictures → Manage Views
Capture views	View → Connect Camera for Views
Connect camera	Ctrl+Shift+C or View → Connect Camera
Hardware panel	Extensions → Open Hardware Connect...
Arm USB sending	Extensions → Send Commands over USB
AprilTag calibration	Hardware Connect → AprilTag Calibration
Models	Settings → Models
Detection	Settings → Detection
Build prompts	Build → Open Build Options...
Gripper presets	Build → Gripper AI...
API key	Settings → API Config → Add manual API key
Restore defaults	Settings → Restore Defaults

From image to action

Clear scene → outlines → task → plan → simulate → calibrate → connect → arm → supervise.

A3-Terra · Humanoid Operating System

Usage Guide · Source-based edition · The homepage uses the supplied authentic launch screenshot.